

Rapid conjugate vertical eye movements, such as those seen in our patient, must be distinguished from vertical nystagmus, such as might be seen in patients with barbiturate intoxication, on discontinuation of tricyclic antidepressants, or associated with paraneoplastic syndromes and demyelinating diseases.⁵ The differential diagnosis of systemic diseases, central nervous system disorders, or possible drug consumption should be considered when evaluating patients with involuntary conjugate eye movements. Formal eye movement recordings may help characterize abnormal eye movements in ambiguous cases.

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Myelin oligodendrocyte glycoprotein antibody-associated bilateral optic neuritis with unilateral prepapillary vitreous hemorrhage

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We present a case of bilateral myelin oligodendrocyte glycoprotein antibody optic neuritis (MOG-ON) with unilateral vitreous hemorrhage in a 6-year-old boy. Multidisciplinary team care, including extensive investigations and prolonged follow-up, excluded alternative causes of vitreous hemorrhage. Early suspicion and diagnosis of MOG-ON led to timely intravenous methylprednisolone treatment, with protracted oral steroid taper, resulting in complete resolution of optic nerve swelling, vitreous hemorrhage, and visual function.

Case Report

A 6-year-old boy presented to Bunbury Regional Hospital with a 3-day history of bilateral blurred vision, left temporal headache and photopsias. There was no recent history of trauma, fever, vaccination, eye pain, nor pain on eye movement. He had no significant ophthalmic, past medical or family history; there was no personal or family history of autoimmune disorders or blood dyscrasias.

He was transferred to Perth Children's Hospital under the care of Neurology and underwent extensive further evaluation, with Ophthalmology consultation. He appeared systemically well, with normal vital signs. Best-corrected visual acuity was counting fingers at 0.5' in each eye. Pupillary reactions were sluggish to light in each eye, with no relative afferent pupillary defect. Anterior segment examination was unremarkable. Careful dilated fundus examination revealed optic disk swelling with peripapillary retinal nerve fiber layer hemorrhages in each eye and a prepapillary vitreous hemorrhage partially obscuring the optic disk of the right eye (Figure 1). There was no evidence of retinal breaks, posterior vitreous detachment, retinal vasculitis, retinitis, vitritis, or neovascularization. Optos fundus imaging (Perth, Western Australia) facilitated peripheral retinal views as scleral indentation could not be performed in this young child without anesthesia. Neurological and systemic examination were unremarkable.

Noncontrast magnetic resonance imaging (MRI) of the brain and orbits demonstrated bilateral long segment hyperintensity of the intraorbital, canalicular, and prechiasmatic segments of both optic nerves, with mild expansion suggestive of edema. Brain parenchyma was normal, with no features of intracranial hemorrhage nor blood products tracking along the optic nerve sheaths (Figure 2). There were no radiological features of raised intracranial pressure. MRI of the spine showed normal spinal cord, with no evidence of demyelination. Although the patient tolerated initial unenhanced MRI sequences without anesthesia, he was unable to endure intravenous contrast administration or prolonged scanning.

Lumbar puncture demonstrated normal opening pressure of 24 cm H₂O, with unremarkable cerebrospinal fluid analysis, including normal differential, protein, and glucose; bacterial culture, angiotensin-converting enzyme, and oligoclonal bands were also normal.

As vitreous hemorrhages are unusual in optic neuritis, extensive investigations included screening for infection,

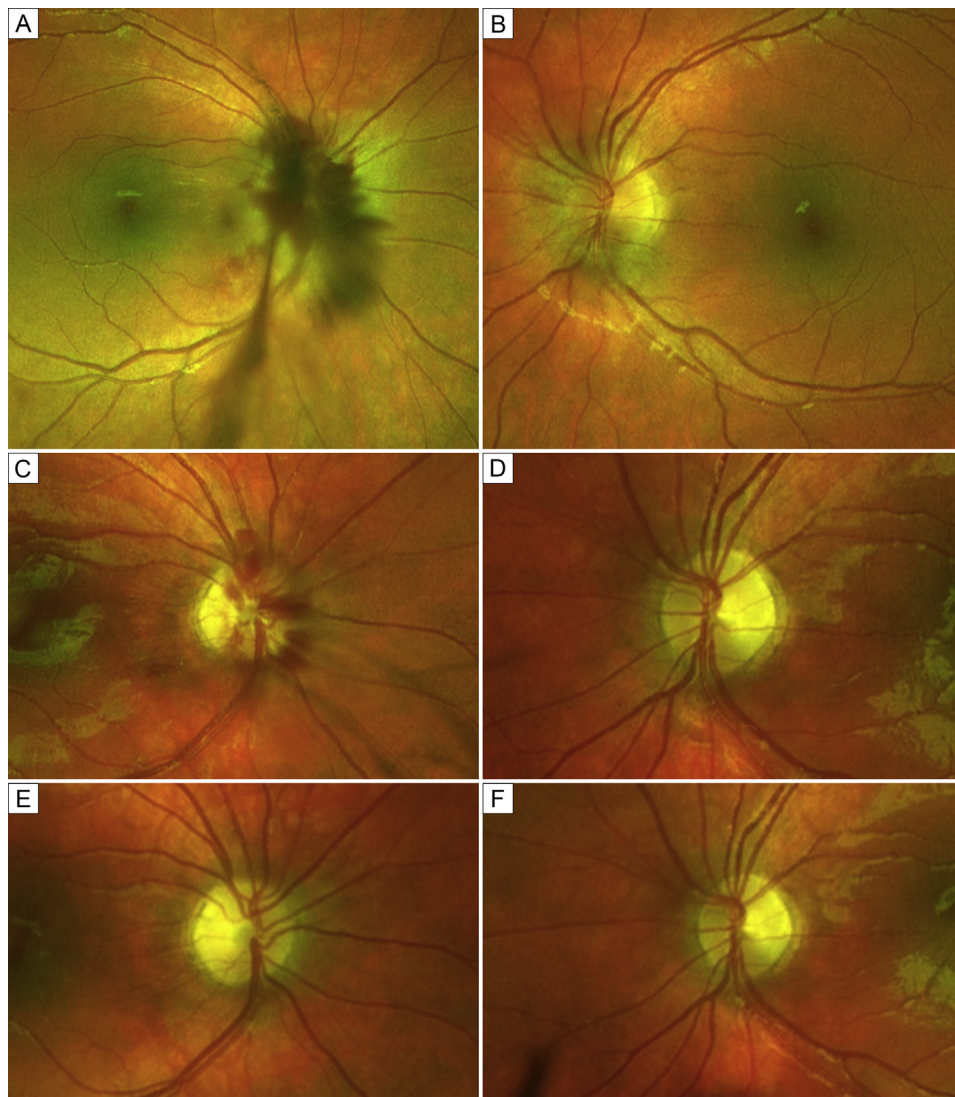


FIG 1. A-B, Optos color fundus photographs centered on the optic disks, at presentation. Prepapillary vitreous hemorrhage with partial obscuration of the swollen optic disk and retinal nerve fiber layer hemorrhages in the right eye (A); mild optic disk swelling with subtle retinal nerve fibre hemorrhages in the left eye (B). C-D, Six weeks after presentation. Resolving prepapillary vitreous hemorrhage in the right eye (C); resolved optic disk edema and temporal pallor in the left eye (D). E-F, Four months after presentation. Resolved vitreous hemorrhage with temporal optic disk pallor, with no neovascularization, vascular loop, or congenital disk anomaly in the right eye (E); temporal optic disk pallor in the left eye (F).

coagulopathy, malignancy, and metabolic and autoimmune diseases, revealing no abnormalities. Laboratory tests included full blood count, blood film, erythrocyte sedimentation rate, C-reactive protein, antinuclear antibody, anti-double stranded DNA, extractable nuclear antigen antibodies, aquaporin-4 antibodies, coagulation profile, factor VIII, factor XIII, and von Willebrand factor, along with serology for Epstein-Barr virus, cytomegalovirus, mycoplasma, rubella, toxoplasma, borreliosis, bartonella, and brucella. Renal ultrasound for vasculitis work-up was normal. MOG antibodies were detected; hence, the patient was diagnosed with bilateral myelin oligodendrocyte glycoprotein antibody optic neuritis (MOG-ON).

The patient was treated with intravenous methylprednisolone (IVMP; 30 mg/kg/day) for 5 days, then oral prednisolone (2 mg/kg/day), which was tapered over 6 weeks.

After IVMP, best-corrected visual acuity improved to 6/30 in the right eye and 6/45 in the left eye. Six weeks after presentation, there was further improvement of visual acuity to 6/6 in each eye, with partial resolution of prepapillary vitreous hemorrhage in the right eye and resolution of optic nerve swelling in each eye (Figure 1). Four months after presentation, the vitreous hemorrhage had completely resolved, allowing clear visualization of the underlying optic disk in the right eye. Temporal pallor was seen, but there were no structural anomalies, including vascular loop, optic disk drusen, or neovascularization (Figure 1). Follow-up MRI at 6 weeks showed persistent but improved low-grade T2 hyperintensity within the intraorbital, intracanalicular and prechiasmatic optic nerve portions, with resolution of optic nerve edema (Figure 2).

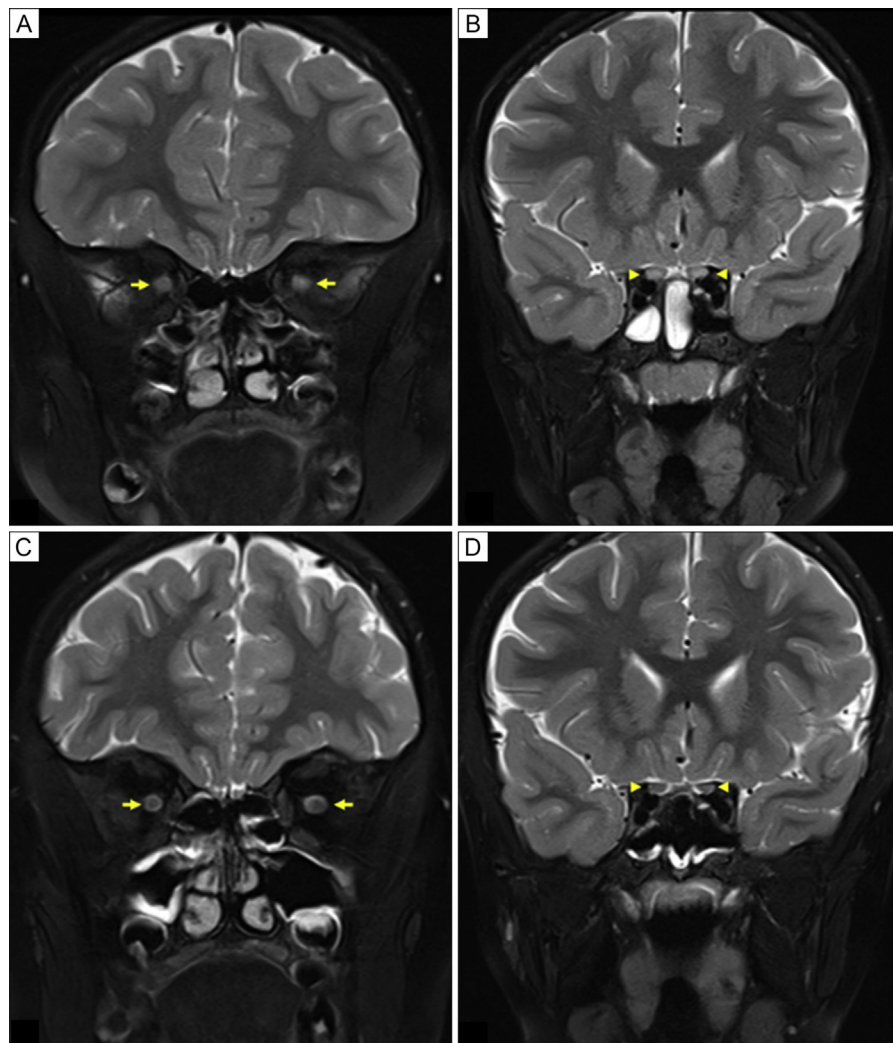


FIG 2. A-B, Magnetic resonance imaging (MRI) at presentation: coronal short-tau inversion recovery (STIR) dataset of the orbits showing mild expansion and increase in STIR signal abnormality of the intraorbital (A, arrows), canalicular (not shown) and prechiasmatic optic nerves (B, arrowhead), consistent with optic nerve edema and optic neuritis. The remainder of the anterior optic pathway was normal. C-D, MRI at 6 weeks after presentation: coronal STIR data set of the orbits showing reduction in edema and signal abnormality of the intraorbital (C, arrows), canalicular (not shown) and prechiasmatic optic nerves (D, arrowheads), in keeping with marked interval improvement.

Discussion

To our knowledge this is the first reported case of vitreous hemorrhage secondary to MOG-ON.

Myelin oligodendrocyte glycoprotein antibody-associated disorder (MOGAD) has varied phenotypic presentations, including optic neuritis, transverse myelitis, acute disseminating encephalomyelitis (ADEM), and brainstem encephalitis.¹ Optic neuritis is the most common initial presentation in all age groups (55%), but ADEM is more common in our patient's demographic (children 4-8 years of age).^{1,2} Pediatric MOG-ON most commonly manifests bilaterally, with accompanying optic disk edema, as demonstrated in our case.²

Dense vitreous hemorrhages in the setting of optic neuritis could potentially obscure optic disk swelling and contribute to diagnostic uncertainty. In this case, optic

disk swelling was obvious despite vitreous hemorrhage, and optic neuritis remained at the top of the differential diagnosis in the context of severe bilateral vision loss that was disproportionate to the ocular findings. MRI was crucial for confirming optic neuritis and excluding other etiologies (eg, subarachnoid hemorrhage).

MOG-ON is often highly steroid responsive, but it is associated with a high relapse rate in approximately 70% of cases due to rapid steroid weaning.^{2,3} This highlights the importance of cautious prednisolone taper and concurrent visual function monitoring.^{2,3}

Pediatric vitreous hemorrhage etiology differs from the adult population and is less well described because of decreased prevalence. The most common causes are trauma, followed by retinopathy of prematurity.⁴ Less common causes include hematologic disorders, retinal

vasculitis, pars plantis, Terson syndrome, retinal tears, diabetic retinopathy, retinal tumors, and retinal vascular disorders, such as familial exudative vitreoretinopathy, Eales disease and Coats disease.⁴⁻⁶ In our case, the vitreous hemorrhage location anterior to the optic disk makes most of these etiologies unlikely, and extensive investigations helped eliminate concurrent pathologies. Although rare, preexisting structural anomalies predisposing to intraocular bleeds, such as a prepapillary vascular loop, were also excluded after vitreous hemorrhage resolution (Figure 1).⁷

Concurrent vitreous hemorrhage in the presence of optic neuritis is rare, with few documented reports. To our knowledge, this is the first reported case of vitreous hemorrhage secondary to confirmed MOG-ON. Chan and colleagues⁸ reported a case of ADEM with optic neuritis and marked peripapillary hemorrhages, although it is unclear whether these hemorrhages extended into the vitreous. It is possible that this case was an example of MOGAD; however, our understanding of MOGAD in 2011 was limited, and MOG antibody testing was not yet widely available.⁸ Vitreous hemorrhage has been reported in cases of neuroretinitis; however, the clinical picture in our case was inconsistent with this diagnosis.^{9,10} Degueldre and Bonnet¹⁰ proposed that vitreous hemorrhage in cat scratch disease could be due to vascular insult caused by Bartonella, resulting in a vascular occlusive phenomenon. This pathophysiology would be mechanistically different to our case. Rather, we postulate the likely mechanism is localized edema at the optic nerve head leading to venous hypertension and rupture of engorged peripapillary retinal vessels. However, many optic neuropathies of varying etiologies cause localized edema of the optic disk but rarely result in vitreous hemorrhage. This suggests that another factor must be at play, for example, a precipitous pressure spike, as seen in Terson's syndrome (though this was not suspected in the present case), or a preexisting breach of the posterior hyaloid face.

Prepapillary vitreous hemorrhage secondary to MOG-ON or other optic neuritis is rare in pediatric patients, and alternative diagnoses must be considered. In cases where vitreous hemorrhage completely obscures the view of the optic disk or where vision loss is disproportionate to the ocular findings, MRI of the anterior visual pathways can be helpful in establishing the diagnosis of optic neuritis. Our case contributes another clinical manifestation to the already broad phenotypic expression of MOGAD in children.

Literature Search

A comprehensive search of the literature was conducted using PubMed, UWA OneSearch, and Google Scholar for English-language results, without date restriction, with the last search completed on the September 30, 2021. Search combinations included *myelin oligodendrocyte glycoprotein antibody optic neuritis* OR *MOG-ON* OR *myelin oligo-*

dendrocyte glycoprotein antibody-associated disorder OR *MOGAD* OR *anti-myelin oligodendrocyte glycoprotein* OR *anti-mog* or *optic neuritis*, and *vitreous haemorrhage*.

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Ophthalmic Findings Associated with *NEDD4L*-related Disorder

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Pathogenic variants in the *NEDD4L* gene are associated with a very rare neurodevelopmental disorder characterized by periventricular

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